

RENANCE

PETROLEUM TRAINING INSTITUTE, EFFURUN

EEI 431 — Past Questions & Materials

by Renance

Course: EEI 431

Institution: Petroleum Training Institute, Effurun

Papers in this pack: 1

Levels: HND1

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Paper 1 of 1 — EEI 431 Assignment

Level HND1 • Second Semester • Handout

EEI 431 ASSIGNMENT

1. In a particular industrial situation, a nicrosil-nisil thermocouple with nicrosil-nisil extension wires is used to measure the temperature of a fluid. In connecting up this measurement system, the instrumentation engineer responsible has inadvertently interchanged the extension wires from the thermocouple (such that the nicrosil thermocouple wire is connected to the nisil extension lead wire, etc.). The open ends of the extension leads are held at a reference temperature of 00C and are connected to a voltmeter, which measures an e.m.f. of 17.51 mV. If the junction between the thermocouple and extension wires is at a temperature of 1400C:

- a. What temperature of fluid is indicated?
- b. What is the true fluid temperature?

2. Resistance thermometers and thermistors are both temperature-measuring devices that convert the measured temperature into a resistance change. What are the main differences between these two types of devices in respect of the materials used in their constructions, their cost, and their operating characteristics?

3. While installing a chromel-constantan thermocouple to measure the temperature of a fluid, it is connected by mistake with copper-constantan extension leads (such that the two constantan wires are connected together and the copper extension wire is connected to the chromel thermocouple wire). If the fluid temperature was actually 250oC and the junction between the thermocouple and extension wires was at 80oC,

- a. what e.m.f. would be measured at the open ends of the extension wires if the reference junction is maintained at 00C?
- b. What fluid temperature would be deduced from this (assuming that the connection mistake was not known)?

4. A copper-constantan thermocouple is connected to copper-constantan extension wires and the reference junction is exposed to a room temperature of 200C. If the output voltage measured is 6.537 mV, what is the indicated temperature at the hot junction of the thermocouple?